

APPLIED AI & ROBOTICS · HOUSTON CITY COLLEGE

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Building Intelligent Systems From NLP Pipelines to Neural Networks

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From Foundations to NLP

Six courses building toward a complete understanding of intelligent systems from AI philosophy to deep learning and natural language processing.



12 NLP modules covering text processing, sentiment analysis, topic modeling, machine translation, and conversational AI

NewsBot 2.0

An NL Ppipeline that evolved from a midterm prototype into a full-featured, multilingual final project.

NewsBot NLP Midterm

Text classification with Naive Bayes and TF-IDF. Sentiment analysis, POS tagging, syntax parsing, and named entity recognition.

- 96.3% accuracy on BBC News dataset
- NLTK, spaCy, scikit-learn, pandas

NewsBot 2.0 NLP Final

Upgraded end-to-end NLP pipeline. Adds LDA topic modeling, LSA summarization, semantic search, multilingual translation, and a conversational chatbot interface.

- 96%+ accuracy · 1,490 articles · 4 languages
- NLTK, spaCy, scikit-learn, sumy, langdetect, deep-translator

96%+ Accuracy

4 Languages

English · German · French · Spanish

[Project Repository](#)

[Live Demo](#)

MIRA: Multi-Agent Event Decor Intelligence System

A rule-based multi-agent AI system built for a luxury event decor company, routing clients and workers to three specialized agents.

Agent Roles

- MIRA Client Q&A, pricing & booking
- MIZEL Creative vision & mood boards
- MILO Worker inventory (passcode protected)

Technical Build

- Rule-based logic with keyword matching
- In-session memory for inventory tracking
- Role-based routing system
- Built in Python / Google Colab

3 Agents

MIRA · MIZEL · MILO

1 System

Client · Creative · Inventory

[Project Repository](#)

[Live Demo](#)

Technical Skills

- 1Python
- 2NLTK
- 3spaCy
- 4scikit-learn
- 5PyTorch
- 6Hugging Face
- 7VS Code
- 8Google Colab
- 9HTML/CSS
- 10CNNs
- 11RNNs
- 12Transformers
- 13Multi-agent Systems
- 14NLP Pipelines
- 15AI Cybersecurity
- 16Database Design

What Makes Me Unique



Multilingual by Nature
Born and raised in Germany, fluent in multiple languages, built multilingual AI because I understand what language means personally.



Design Meets AI
Trained in graphic design and 3D design at a vocational college in Germany I build systems that work and look professional.



Security Aware
Studied AI Cybersecurity I understand the risks behind the systems I build, not just how to build them.

Looking Back, Moving Forward

What I Learned

- How to build real AI systems from scratch
- The full ML workflow end-to-end
- How chatbots, translators, and agents work under the hood
- The ethics and security side of AI

Where I'm Headed

- Build a web app for NewsBot 2.0
- Explore BERT and transformer models deeper
- Apply AI skills professionally
- Keep growing this portfolio